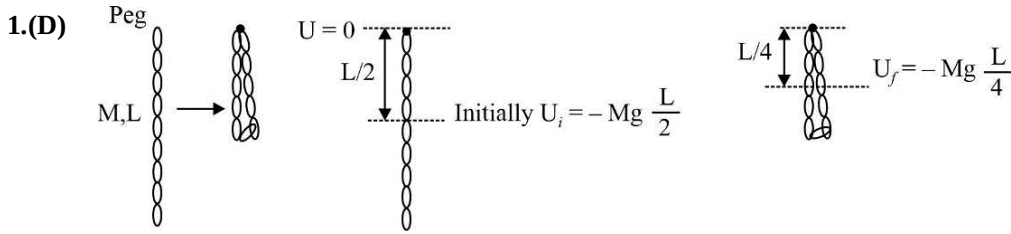


Solutions to JEE MAIN – 4 | JEE - 2022

PHYSICS

SECTION-1



Work done by external agent = Gain in Gravitational potential energy

$$= U_f - U_i = -\frac{MgL}{4} - \left(-\frac{MgL}{2}\right) = \frac{MgL}{4}$$

2.(D) Before the string is cut, tension in the spring = $3mg$

Tension in spring doesn't change instantly.

So, just after the string is cut, tension in spring is still $3mg$

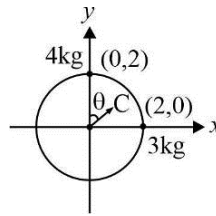
$$\therefore a_A = \frac{3mg - mg}{m} = 2g \text{ upwards}$$

3.(B) $X_{cm} = \frac{3 \times 2 + 4 \times 0 + 8 \times 0}{3 + 4 + 8} = \frac{6}{15}$

$$Y_{cm} = \frac{3 \times 0 + 4 \times 2 + 8 \times 0}{3 + 4 + 8} = \frac{8}{15}$$

$$\tan \theta = \frac{X_{cm}}{Y_{cm}} = \frac{3}{4}$$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right)$$



4.(B) Loss in PE = Gain in KE

$$\frac{1}{2}(k + 3k + 2k + 4k) \left(x^2 - \left(\frac{x}{2}\right)^2 \right) = \frac{1}{2}mv^2$$

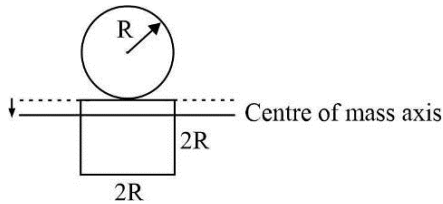
$$\frac{1}{2}(10k) \left(\frac{3x^2}{4} \right) = \frac{1}{2}mv^2$$

$$v = x \sqrt{\frac{15k}{2m}}$$

5.(A) $K = K_T + K_R = \frac{1}{2}(3m)(\omega R)^2 + \frac{1}{2} \left[\frac{1}{2}mR^2 + \frac{2m(2R)^2}{12} \right] \omega^2$

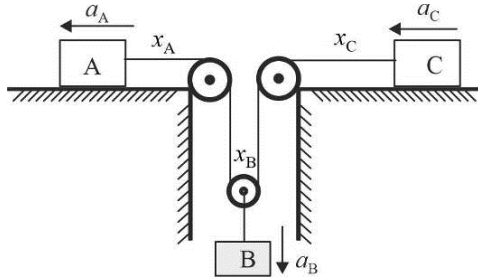
$$= \frac{25}{12}m\omega^2 R^2$$

6.(D)



I_{cm} will be minimum therefore I will first decrease then increase.

7.(A)



Length of the string is $l = x_A + 2x_B + x_C$

Differentiating twice w.r.t. time, we get $0 = \frac{d^2 x_A}{dt^2} + \frac{2d^2 x_B}{dt^2} + \frac{d^2 x_C}{dt^2}$

$$0 = a_A + 2a_B - a_C$$

8.(A) $0 \leq t \leq 2 \rightarrow a = 0 \{F \leq f_L = 16N\}$

$$\text{For, } t > 2 \quad a = \frac{4t^2 - 16}{4} = t^2 - 4$$

$$\frac{dv}{dt} = t^2 - 4$$

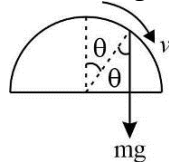
$$\int_0^v dv = \int_2^5 (t^2 - 4) dt \Rightarrow v = 27 \text{ m/s}$$

9.(D) $x = 10t^2 + 20$

$$v = \frac{dx}{dt} = 20t$$

$$W = \Delta KE = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \frac{1}{2}(1.5)(40)^2 = 1200 \text{ J}$$

10.(A) At the time of leaving contact



$$N = 0$$

$$\therefore mg \cos \theta = \frac{mv^2}{R} = \frac{m(2gh)}{R} \therefore \cos \theta = \frac{2h}{R} = \frac{2 \left[\frac{R}{5} + R(1 - \cos \theta) \right]}{R}$$

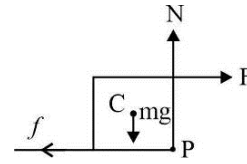
On solving this equation, we get $\cos \theta = 4/5$ or $\theta = \cos^{-1} \left(\frac{4}{5} \right)$

11.(D) $K_i + U_i = K_f + U_f \therefore 0 + 5 = \frac{1}{2} \times 10 \times v^2 + 1 \therefore v = \frac{2}{\sqrt{5}} m/s$

12.(C) At the instant of toppling N will pass through the corner edge: and taking torque about P, we have:

$$\left(mg \times \frac{a}{2} \right) = (F \times a)$$

$$F = \frac{mg}{2} = \frac{1 \times 10}{2} = 5N \Rightarrow \frac{5t}{4} = 5 \Rightarrow t = 4 \text{ second}$$



13.(A) At, B total kinetic energy = mgh

Here, $m = \text{mass of ball}$

The ratio of rotational to translational kinetic energy would be,

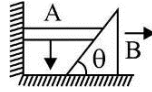
$$\frac{K_R}{K_T} = \frac{2}{5} \Rightarrow \therefore K_R = \frac{2}{7} mgh \text{ and } K_T = \frac{5}{7} mgh$$

In portion BC, friction is absent. Therefore, rotational kinetic energy will remain constant and translational kinetic energy will convert into potential energy. Hence, if H be the height to which ball climbs in BC, then

$$mgH = K_T \text{ or } mgH = \frac{5}{7} mgh \text{ or } H = \frac{5}{7} h$$

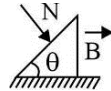
14.(C) $a_A = a_B \tan \theta$

$$\Rightarrow \tan \theta = \frac{a_A}{a_B} = 2$$



$$N \sin \theta = m_B \cdot a_B$$

$$N = \frac{m_B \cdot a_B}{\sin \theta} = \frac{\sqrt{5}}{2}$$



15.(B) By linear momentum conservation.

Along x-axis (No external force in horizontal direction):

$$3m \times (20) = m \times (0) + m \times (0) + mv_x$$

$$v_x = 60 ms^{-1}$$

Along y-axis (weight is external for system but it is non-impulsive):

$$0 = m \times (80) + m \times (0) + mv_y$$

$$v_y = -80 ms^{-1}$$

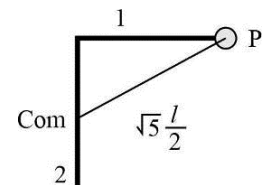
$$\text{Speed of C} = \sqrt{V_x^2 + V_y^2} = \sqrt{60^2 + 80^2} = 100 ms^{-1}$$

16.(B) Moment of inertia of rod 1 about the axis, $I_1 = \frac{ml^2}{3}$

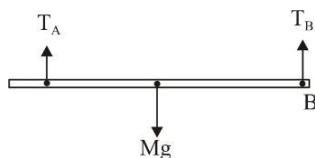
$$\text{Moment of inertia of rod 2 about the axis, } I_2 = \frac{ml^2}{12} + m \left(\sqrt{5} \frac{l}{2} \right)^2$$

Moment of Inertia of the system, $I = I_1 + I_2$

$$I = \frac{ml^2}{3} + \frac{ml^2}{12} + m \left(\frac{\sqrt{5}l}{2} \right)^2 = \frac{5ml^2}{3}$$



$$17.(D) \sum \tau_B = T_A \left(\frac{5L}{6} \right) - Mg(L/2) = 0$$



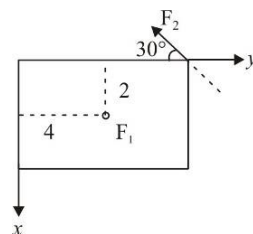
Therefore

$$T_A = (MgL/2) / (5L/6) = (3Mg/5)$$

$$18.(C) \vec{\tau}_1 = (2\hat{i} + 4\hat{j}) \times (-\hat{k}) F = (2\hat{j} - 4\hat{i}) F$$

$$\vec{\tau}_2 = (8\hat{j}) \times \left(\frac{F}{2} \right) (-\hat{i}) = 4F\hat{k}$$

$$\text{So, } \vec{\tau} = (-4\hat{i} + 2\hat{j} + 4\hat{k}) F$$



$$19.(D) I = MK^2$$

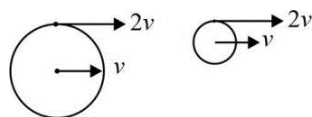
$$(A) \text{ Ring} \rightarrow MR^2 = MK^2 \Rightarrow K = R$$

$$(B) \text{ Disc} \rightarrow \frac{1}{2} MR^2 = MK^2 \Rightarrow K = R\sqrt{\frac{1}{2}} = R\sqrt{0.5}$$

$$(C) \text{ Hollow Sphere} \rightarrow \frac{2}{3} MR^2 = MK^2 \Rightarrow K = R\sqrt{\frac{2}{3}} = R\sqrt{0.66}$$

$$(D) \text{ Solid Sphere} \rightarrow \frac{2}{5} MR^2 = MK^2 \Rightarrow K = R\sqrt{\frac{2}{5}} = R\sqrt{0.4}$$

$$20.(A) \text{ Velocity of the moving plank} = 2v$$



Therefore, velocity of the centre of the large cylinder will be v.

SECTION-2

$$21.(0.75)$$

The acceleration of the box till the slipping continues is

$$a = \frac{f_k}{m} \text{ or } a = \frac{\mu mg}{m} = \mu g = (0.6)(10) = 6 \text{ m/sec}^2$$

Slipping will continue till its velocity also becomes 3 m/sec

$$\text{Therefore, } v = u + at \text{ or } 3 = 0 + 6t \text{ or } t = 0.5 \text{ sec}$$

$$\text{In this time, displacement of the box is } s_1 = \frac{1}{2} at^2 = \frac{1}{2} (6)(0.5)^2 = 0.75 \text{ m}$$

$$\text{And displacement of the belt is } s_2 = ut = (3)(0.5) = 1.5 \text{ m}$$

$$\text{Therefore, displacement of the box with respect to the belt is } |s_1 - s_2| = 0.75 \text{ m}$$

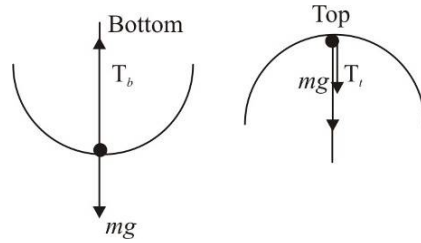
$$22.(12) \text{ Loss in GPE} = \text{gain in K.E.}$$

$$mgh = \frac{1}{2} mv^2 + \frac{1}{2} \left(\frac{2}{3} mr^2 \right) \left(\frac{v}{r} \right)^2 = \frac{5}{6} mv^2 \Rightarrow v = \sqrt{\frac{6gh}{5}}$$

$$v = \sqrt{\frac{6 \times 10 \times 12}{5}} = 12 \text{ m/s}$$

$$23.(7) \text{ Let } u \text{ and } v \text{ be the speeds of sphere at the bottom and the top positions and } m \text{ be the mass.}$$

$$\text{Radius of circle} = \text{length of string} = r = 0.7 \text{ m}$$



T_b = tension at the bottom or the maximum tension

T_t = tension at the top or the minimum tension

$$T_b - mg = \frac{mu^2}{r}; \quad T_t + mg = \frac{mv^2}{r}$$

$$T_b = 4T_t$$

$$\left(\frac{mu^2}{r} + mg \right) = 4 \left(\frac{mv^2}{r} - mg \right)$$

$$(4v^2 - u^2) = 5rg \dots\dots(i)$$

Using conservation of energy,

Loss in KE from bottom to the top = Gain in GPE.

$$\frac{1}{2}mu^2 - \frac{1}{2}mv^2 = mg(2r)$$

$$v^2 = u^2 - 4rg \dots\dots(ii)$$

Using (i) and (ii), we get $4(u^2 - 4rg) - u^2 = 5rg$

$$3u^2 = 21rg$$

$$u = \sqrt{7rg} = \sqrt{7(0.7)10} = 7 \text{ m/s}$$

24.(40) If angular acceleration of the cylinder is α and acceleration of C.O.M. is a , then

$$60 - f = ma \dots\dots(i)$$

$$(60 + f)R = I_{cm}\alpha \dots\dots(ii)$$

$$a = R\alpha \dots\dots(iii)$$

From (ii) and (iii)

$$60 + f = I_{cm} \frac{a}{R^2} \dots\dots(iv)$$

Adding (i) and (iv)

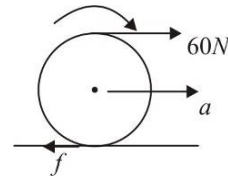
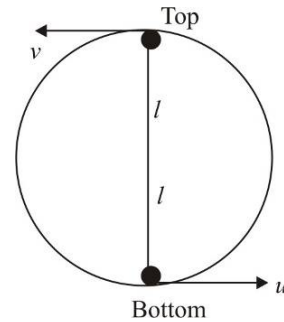
$$120 = \left(\frac{I_{cm}}{R^2} + m \right) a; \quad a = \frac{120}{\left(m + \frac{mR^2}{2R^2} \right)} = \frac{80}{m}$$

$$\alpha = \frac{80}{mR} \text{ or } \alpha = \frac{80}{4 \times 0.5}; \quad \alpha = 40 \text{ rad/s}^2$$

25.(10) By Angular momentum conservation:

$$\left(\frac{1}{2}MR^2 \right) \omega = \left(\frac{1}{2}MR^2 + (\mu t)R^2 \right) \frac{\omega}{4}$$

$$\Rightarrow \quad \frac{M}{2} = \left(\frac{M}{2} + \mu t \right) \frac{1}{4} \Rightarrow \quad t = \frac{3M}{2\mu} = \frac{1.5M}{\mu} = \frac{15M}{10\mu} \Rightarrow \quad K = 10$$



CHEMISTRY

SECTION-1

1.(D) $\frac{1}{3} \times \text{Maximum capacity of the balloon} = 150 \text{ mL}$

So maximum capacity of the balloon $= 150 \times 3 \text{ mL} = 450 \text{ mL}$

$v_1 = 150 \text{ mL}$

$T_1 = 7 + 273 \text{ K} = 280 \text{ K}$

$n = 1 \text{ mol}$

The balloon will burst at temperature T_2 when volume becomes 450 mL

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{150}{280} = \frac{450}{T_2}$$

$T_2 = 840 \text{ K}$ or 567°C

So, the balloon will not burst when temperature is 300°C

But will burst at temperature 567°C and above

2.(C) $n_A = \frac{30}{60} = 0.5 \text{ mol}$

$n_B = \frac{90}{45} = 2 \text{ mol}$

$P_T \times x_A = p_A$

$100 \times \frac{0.5}{2.5} = p_A$

$20 \text{ mm Hg} = p_A$

3.(B) $Z = \frac{PV}{nRT} = \frac{3 \times 5}{1 \times 0.08 \times 600} = \frac{15}{48} = \frac{5}{16}$

Since $Z < 1$, so gas shows negative deviation and volume occupied is less than that occupied by ideal gas

4.(B) (I) $T_1 > T_2 > T_3$ (II) $P_4 < P_3 < P_2 < P_1$

5.(C) $P = \frac{dRT}{M_0}$

$$\frac{P_A}{P_B} \times \frac{M_{0(A)}}{M_{0(B)}} = \frac{d_A}{d_B}$$

$$\frac{P_A}{P_B} \times 4 = \frac{1}{2}$$

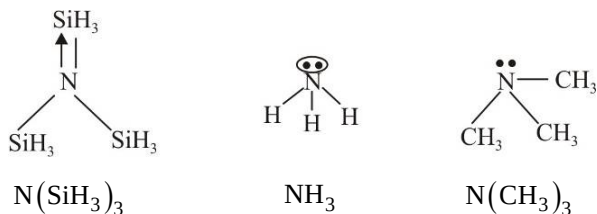
$$\frac{P_A}{P_B} = \frac{1}{8}$$

6.(B) $T_b = \frac{a}{Rb}, T_c = \frac{8a}{27Rb}, P_c = \frac{a}{27b^2}$

7.(C) II Statement is wrong.

8.(B) SnCl_4 has more covalent character than SnCl_2 .

9.(C)

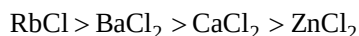


In $\text{N}(\text{SiH}_3)_3$, there is backbonding from $\text{N} \rightarrow \text{Si}$ which changes hybridisation from sp^3 to sp^2 so bond angle is 120° .

In NH_3 , because of the lone pair on nitrogen, the bond angle is less than 109.5° .

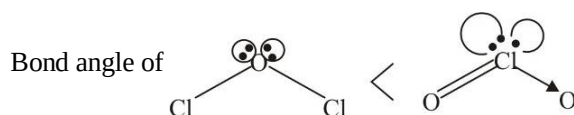
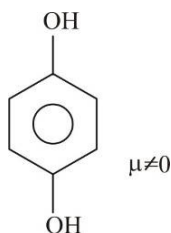
In $\text{N}(\text{CH}_3)_3$, because of the bulky $-\text{CH}_3$ group, the bond angle is more than 109.5° .

10.(C) Order of ionic character:

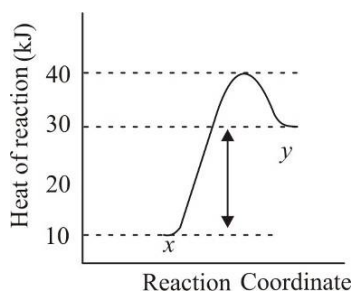


11.(D) Bond length is minimum in BF_3 due to most effective back bonding.

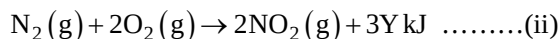
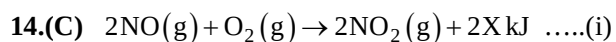
12.(B) OF_4 and OF_6 do not exist as oxygen does not have vacant d - orbitals in its valence shell



13.(C)



$$\Delta H_{x \rightarrow y} = 30 - 10 = 20 \text{ kJ}$$



The enthalpy of formation of NO is $= \frac{2X}{2} - \frac{3Y}{2} = X - \frac{3Y}{2}$

15.(B) Enthalpy for the reduction of 3 double bonds of $C_6H_6(l)$ = Enthalpy of hydrogenation + Enthalpy of resonance = $-141 + (-225) = -366 \text{ kJ/mol}$

$$\text{Enthalpy for hydrogenation of 1,4 cyclohexadiene} = -\frac{366}{3} \times 2 = -244 \text{ kJ/mol}$$

16.(A) $\Delta H = 6B.E_{C-H} + B.E_{C-C} + B.E_{Cl-Cl} - (5B.E_{C-H} + B.E_{C-C} + B.E_{C-Cl} + B.E_{H-Cl})$

$$-50 = B.E_{C-H} + B.E_{Cl-Cl} - B.E_{C-Cl} - B.E_{H-Cl}$$

$$-50 = a + b - 84 - 103$$

$$a + b = 187 - 50 = 137 \dots\dots(i)$$

$$\frac{a}{b} = \frac{9}{5}$$

$$a = \frac{9}{5}b \dots\dots(ii)$$

Putting value of a in equation (i)

$$a + b = 137$$

$$\frac{9}{5}b + b = 137$$

$$\frac{14}{5}b = 137$$

$$b = \frac{137 \times 5}{14} = 48.9 \approx 49$$

17.(D) $NH_4OH_{(aq)} + HCl_{(aq)} \rightarrow NH_4Cl_{(aq)} + H_2O(l)$; $\Delta_r H = -50 \text{ kJ/mol}$

$$\text{Enthalpy of ionisation of } NH_4OH = (-50 + 57) \text{ kJ/mol} = 7 \text{ kJ/mol}$$

18.(C) PH_5 do not exist because hydrogen will not be able to excite electron of phosphorus in vacant 3d orbitals to create the availability of 5 unpaired electron.

19.(B) $C_{12}H_{22}O_{11} + 12O_2 \rightarrow 12CO_2 + 11H_2O$; $\Delta H = -5600 \text{ kJ}$

$$\text{Mole of } O_2 = \frac{320}{32} = 10 \text{ moles}$$

$$\Delta H = -5600 \times \frac{10}{12} = -4666.67 \text{ kJ}$$

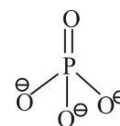
20.(D) Theory Based

SECTION-2

21.(3) NO, O_2, B_2 are paramagnetic

22.(1.25) Bond order $= 1 + \frac{1}{4} = \frac{5}{4} = 1.25$

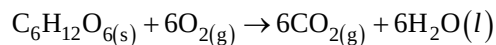
This molecule will undergo resonance to have a bond order of 1.25



$$23.(3) \quad q_v = 600 \times 10 = 6000 \text{ J}$$

For 0.36 gm

$$\text{For 1 mole} = 180 \text{ gm, } \Delta U = -6000 \times \frac{180}{0.36} = -3 \times 10^6 \text{ J / mol}$$



$$\Delta n_g = 0 \Rightarrow \Delta H = \Delta U = -3 \times 10^6 \text{ J / mole}$$

$$x = 3$$

24.(149) For gaseous mixture 80% gas A and 20% gas B:

$$\text{Average molar mass of mixture } (M_{\text{mix}}) = \frac{(0.8 \times 64) + (0.2 \times M_0)}{1}$$

$$\frac{r_A}{r_B} = \sqrt{\frac{M_{\text{mix}}}{64}}$$

$$\frac{1}{\frac{8}{9}} = \sqrt{\frac{M_{\text{mix}}}{64}}$$

$$\frac{9}{8} \times \frac{9}{8} = \frac{M_{\text{mix}}}{64}$$

$$M_{\text{mix}} = 81$$

$$\text{So } 81 = \frac{(0.8 \times 64) + (0.2 \times M_0)}{1}$$

$$149 = M_0$$

$$25.(250) \quad C_{\text{av}} = \sqrt{\frac{8RT}{\pi M_0}}$$

$$\frac{C_{\text{av}_1}}{C_{\text{av}_2}} = \sqrt{\frac{T_1}{T_2}}$$

$$\frac{500}{C_{\text{av}_2}} = \sqrt{\frac{1207 + 273}{97 + 273}} = \sqrt{\frac{1480}{370}} = 2$$

$$C_{\text{av}_2} = \frac{500}{2} = 250 \text{ m / s}$$

MATHEMATICS

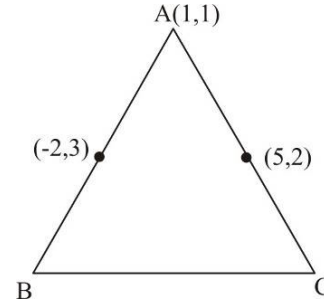
SECTION-1

1.(A) $(-2, 3)$ is mid point of $AB \Rightarrow B = (-5, 5)$

$(5, 2)$ is mid point of $AC \Rightarrow C = (9, 3)$

Centroid of $ABC = \left(\frac{1-5+9}{3}, \frac{1+5+3}{3} \right) = \left(\frac{5}{3}, 3 \right)$

2.(D) Required ratio $= -\left(\frac{3(-2)+4(1)-7}{3(1)+4(2)-7} \right) = -\left(\frac{-9}{4} \right) = 9:4$



3.(D) JANLOKPAL

JNO

OJN

NJO

ONJ

$${}^9C_3 \times 4 \times \frac{6!}{2!2!} = \frac{9 \times 8 \times 7}{6} \times \frac{4 \times 6!}{2!2!} = 12 \times 7!$$

Or JNO can be permuted in $3!$ Ways out of which 4 ways are favourable.

4.(C) $\left(x + \frac{1}{x} \right)^{10} \left(x - \frac{1}{x} \right)^9 \Rightarrow \left(x + \frac{1}{x} \right) \left(x^2 - \frac{1}{x^2} \right)^9$

General term of $\left(x^2 - \frac{1}{x^2} \right)^9 = {}^9C_r x^{18-2r} \left(\frac{1}{x^2} \right)^r (-1)^r = {}^9C_r x^{18-4r} (-1)^r$

Coefficient of $x^5 = -{}^9C_3 = \frac{-9 \times 8 \times 7}{6} = -84$

5.(A) Family of the passing through $(-2, 0)$ hence required line will be $\perp^r$ line joining $(-2, 0)$ and $(2, 3)$

$\Rightarrow$ Slope $= \frac{3}{4}$

Equation of line : $(y-0) = -\frac{4}{3}(x+2) \Rightarrow 3y+4x+8=0$

6.(C) $\sum_{r=0}^{20} {}^{20}C_r \left(-\frac{1}{3} \right)^r = \left(1 - \frac{1}{3} \right)^{20} = \left(\frac{2}{3} \right)^{20} = \frac{2^{20}}{3^{20}}$

7.(D) Both of diagonals of a square are perpendicular hence let the diagonal is $x+7y+k=0$, it passes through

$(-4, 5) \Rightarrow -4+35+k=0$

$k=-31$

Diagonals is $x+y-31=0$

8.(C) Image of (α, β) in line $y=-x$ is $(-\beta, -\alpha)$

9.(A) Equal intercept but opposite in sign $\Rightarrow m=1$

$$\Rightarrow \text{line } x - y = -5 \Rightarrow x + 5 - y = 0$$

$$10.(B) \quad T_{r+1} = {}^9C_r \left(\frac{x^2}{2} \right)^{9-r} \left(-\frac{1}{x} \right)^r = {}^9C_r \frac{x^{18-3r}}{2^{9-r}} (-1)^r$$

$$18 - 3r = 0 \Rightarrow r = 6$$

$$= \frac{{}^9C_6}{2^3} = \frac{9 \times 8 \times 7}{6 \times 8} \Rightarrow k = \frac{21}{2}$$

$$11.(C) \quad 2 \left[{}^6C_0 x^6 + {}^6C_2 x^4 (x^2 - 1) + {}^6C_4 x^2 (x^2 - 1)^2 + {}^6C_6 (x^2 - 1)^3 \right]$$

$$\text{Coefficient of } x^6 = 2 \left[{}^6C_0 + {}^6C_2 + {}^6C_4 + {}^6C_6 \right] = 2[1 + 15 + 15 + 1] = 64$$

$$12.(B) \quad B_1 + B_2 + B_3 + B_4 + B_5 = 7$$

$$\text{Total number of ways} = {}^{11}C_4$$

$$13.(C) \quad 2p = \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{4p^2} = \frac{2}{8p^2}$$

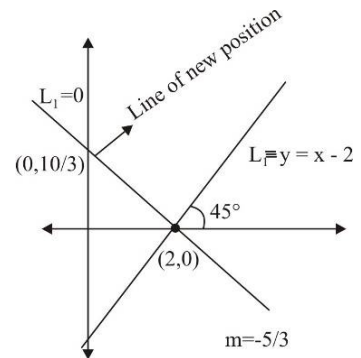
$$\frac{1}{a^2}, \frac{1}{8p^2}, \frac{1}{b^2} \text{ are in A.P.}$$

$$14.(C) \quad \text{Equation of line in new position } y - 0 = -\frac{5}{3}(x - 2) \Rightarrow 3y = -5x + 10$$

$$5x + 3y = 10$$

Area of triangle made by line with co-ordinate

$$\text{Area} = \frac{1}{2} \times 2 \times \frac{10}{3} = \frac{10}{3} \text{ square units}$$



$$15.(B) \quad (1 + x^2)^3 (1 + x^4)^7 (1 - x)^3$$

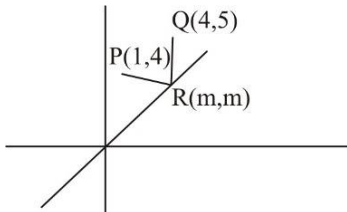
$$= (1 - x)^3 (1 + x^2)^3 (1 + 7x^4 + {}^7C_2 x^8 + \dots)$$

$$\text{Coefficient of } x^5 =$$

$$-{}^3C_3 \times {}^3C_1 - {}^3C_1 \times {}^3C_2 - {}^3C_1 \times 7$$

$$-3 - 9 - 3 \times 7 = -33$$

16.(A)



Take reflection of Q in the line $y = x$. Let it be $Q \equiv (5, 4)$ so, the equation of line PQ is $y = 4$. Now, r must be the point of intersection of line PQ and $y = x$. $m = 4$

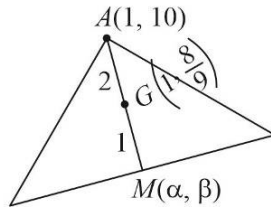
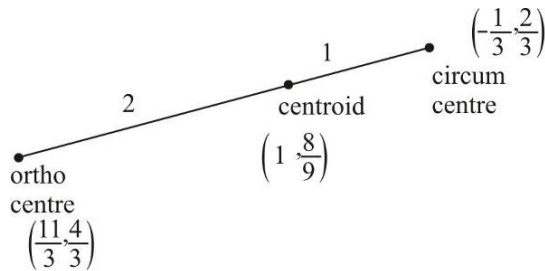
$$17.(C) \quad {}^{10}C_0 - {}^{10}C_1 \frac{2^1}{2} + \frac{{}^{10}C_2 2^2}{3} - \frac{{}^{10}C_3 2^3}{4} + \dots + \frac{{}^{10}C_{10} 2^{10}}{11}$$

$$\int_0^2 (1-x)^{10} dx = \left[\frac{(1-x)^{11}}{-11} \right]_0^2 = \left(+\frac{1}{11} + \frac{1}{11} \right) = \frac{2}{11}$$

$$18.(C) \quad \frac{(27)^{40}}{12} = \frac{3^{120}}{4 \times 3} = \frac{3^{119}}{4} = \frac{(4-1)^{119}}{4} = \frac{(4k-1)3}{4 \times 3} = \frac{12k-3}{12}$$

Remainder = '9'

19.(A)



$$\frac{2\alpha+1}{3} = 1 \quad \frac{2\beta+10}{3} = \frac{8}{9}$$

$$\alpha = 1, \quad \beta = \frac{4}{5} - 5 \Rightarrow \beta = \frac{-11}{5}$$

20.(B) Let third vertex is $C(h, k)$

$$AB \perp CD$$

$$\frac{k-2}{h-1} \times \frac{5-(-3)}{-2-4} = -1$$

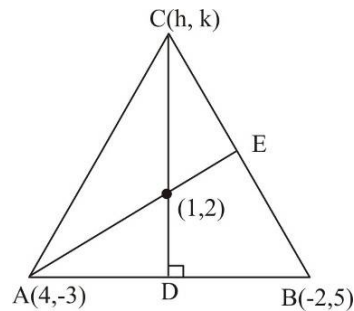
$$3h - 4k + 5 = 0$$

Similarly $AE \perp BC$

$$\frac{-2-2}{4-1} \times \frac{k-5}{h-(-2)} = -1$$

$$3h - 5k + 31 = 0 \dots\dots(ii)$$

Solving both gives $h = 33, k = 26$



SECTION-2

$$21.(972) \quad Ar = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{\sqrt{3}}{4} \times 36 \Rightarrow \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = 972$$

$$22.(2) \quad \left(\frac{1}{x} + x \sin x \right)^{10}$$

$$\text{Middle term} = {}^{10}C_5 \left(\frac{1}{x} \right)^5 (x \sin x)^5 = 7 \frac{7}{8} = \frac{10 \times 9 \times 8 \times 7 \times 6}{120} \times (\sin x)^5 = \frac{63}{8}$$

$$= (\sin x)^5 = \frac{1}{2^5} \Rightarrow \sin x = \frac{1}{2}$$

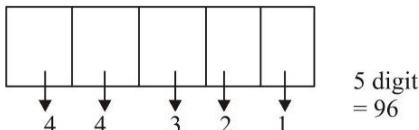
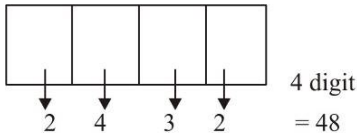
$$23.(3) \quad \left(x^p + \frac{1}{x^2} \right)^{25} \quad T_{r+1} = {}^{25}C_r (x^p)^{25-r} \frac{1}{x^{2r}}$$

$$\text{Coefficient} = 2300 = {}^{25}C_r \Rightarrow r = 3 \text{ or } 22$$

$$22p - 6 = 60 \Rightarrow p = 3$$

$$\text{Or } 3p - 44 = 60 \Rightarrow p = \frac{104}{3} \quad \text{Not possible}$$

$$24.(144)$$



$$\text{Total} = 96 + 48 = 144$$

$$25.(23) \quad 3072 = 2^{10} \times 3$$

$$4 \times 4 \times 4 \times 4 \times 4 \times 3 \times 1 \times 1 \rightarrow \frac{8!}{5! \times 2!} = {}^8C_5 \times 3$$

$$4 \times 4 \times 4 \times 4 \times 2 \times 2 \times 3 \times 1 \rightarrow \frac{8!}{4! \times 2!} = {}^8C_5 \times 5 \times 3$$

$$4 \times 4 \times 4 \times 2 \times 2 \times 2 \times 2 \times 3 \rightarrow \frac{8!}{3! \times 4!} = {}^8C_5 \times 5 = 23 \cdot {}^8C_5$$